# Milestone 3: Baseline Reproduction - Technical Memo

Team ML Musketeers (Group 21) - CSCI 3052U, Fall 2026 - github.com/aurodeeps/student-dropout-prediction, release v0.3-baseline

**Summary.** We built a one-command pipeline that regenerates our baseline table from the approved CSV. Logistic regression reaches macro-F1 **0.651** on the early feature set (enrollment + semester 1) and **0.705** on the full set (+ semester 2) on the validation set, against 0.222-0.313 for the naive benchmarks. All eight benchmark and baseline results from our Milestone 2 memo were reproduced to three decimals. The weak spot is the Enrolled class (recall 0.32 early). The test set has not been used. The group's exploratory notebook (crosstabs by gender, scholarship, tuition, debtor, displacement and attendance) motivated the subgroup check in Section 4.

## 1. Pipeline and reproducibility

| | |
|---|---|
| Data | `data/data.csv` (UCI / Zenodo, CC BY 4.0): 4,424 rows x 37 columns, no missing values. MD5 `9b2b50f41d0cefbbc3e8222d60032d91` (line endings normalised) is checked at every run. |
| Split | Stratified 70/15/15, seed 42: 3,096 train / 664 validation / 664 test. Indices stored in `data/split_indices.json` and compared with the regenerated split each run. **Test set is not evaluated in this milestone.** |
| Preprocessing | 9 integer-coded categorical variables one-hot encoded, other features standardized, inside a scikit-learn `Pipeline` fit on training data only. Feature sets: **early** = 30 features (no semester-2 columns), **full** = 36. |
| Models | Majority class; stratified random (seed 42); multinomial logistic regression (L2, C=1, lbfgs, max_iter 3000; C also searched over {0.01, 0.1, 1, 10, 100} by 5-fold CV on train, C=1 chosen for both sets); decision tree (depth 5, seed 42). |
| Evaluation | Primary metric macro-F1 (fixed in Milestone 2); also accuracy and per-class recall. Validation set plus 5-fold x 3-repeat stratified CV on the training set (seed 42). |
| Environment | Python 3.12.3, scikit-learn 1.8.0, pandas 3.0.2, numpy 2.4.4, scipy 1.17.1 (pinned in `requirements.txt`). Run time about 30 s. |
| Commands | `pip install -r requirements.txt`, then `python scripts/run_baseline.py` (regenerates `results/` and `figures/`); `python -m pytest -q` runs the checks. |

## 2. Baseline table (validation set; CV on training set)

| Features | Model | Macro-F1 | Accuracy | Recall Dropout | Recall Enrolled | CV macro-F1 (mean +/- sd) |
|---|---|---|---|---|---|---|
| Early (30) | Majority class | 0.222 | 0.500 | 0.000 | 0.000 | 0.222 +/- 0.000 |
| Early (30) | Stratified random (seed 42) | 0.313 | 0.373 | 0.319 | 0.126 | 0.333 +/- 0.020 |
| Early (30) | **Logistic regression (C=1)** | **0.651** | 0.736 | 0.714 | 0.319 | 0.659 +/- 0.011 |
| Early (30) | Decision tree (depth 5) | 0.549 | 0.703 | 0.690 | 0.084 | 0.576 +/- 0.026 |
| Full (36) | Majority class | 0.222 | 0.500 | 0.000 | 0.000 | 0.222 +/- 0.000 |
| Full (36) | Stratified random (seed 42) | 0.313 | 0.373 | 0.319 | 0.126 | 0.333 +/- 0.020 |
| Full (36) | **Logistic regression (C=1)** | **0.705** | 0.773 | 0.751 | 0.420 | 0.697 +/- 0.015 |
| Full (36) | Decision tree (depth 5) | 0.602 | 0.720 | 0.634 | 0.193 | 0.639 +/- 0.016 |

## 3. Reproduction check and discrepancy analysis

**Expected values.** The dataset descriptor (Realinho et al., 2022) reports no classification results, so there is no published baseline to reproduce directly, and we do not copy numbers from other studies. We reproduce the reference values in our own Milestone 2 memo with the new pipeline: all 8 benchmark/baseline rows match in macro-F1 and accuracy to 3 decimals (`results/reproduction_check.csv`). The only published context we found is Vo et al. (2026), who report a best macro-F1 of 0.7082 (LightGBM) and best accuracy of 0.7797 (random forest) on this dataset. We read only their abstract and have not checked their split, features or tuning, so we treat it as context, not a target. Our untuned logistic regression on the full set (0.705 / 0.773) is in the same range.

**Discrepancies we found and what explains them** (`results/discrepancy_probes.csv`):

- **Dataset version.** The descriptor lists 35 attributes and numbers courses 1-17. The UCI CSV we use has 36 features (it adds *Admission grade* and *Previous qualification (grade)*), uses course codes such as 171 and 9254, and spells *Nacionality*. Published feature counts are therefore not directly comparable to ours.
- **The naive benchmark depends on the seed.** Stratified random scores 0.313 at seed 42 but averages 0.337 over seeds 0-9 (range 0.316-0.372). Logistic regression (early) beats the seed-42 value by 0.338 and the 10-seed mean by 0.314, both meeting our >= 0.30 success criterion, but beats the luckiest seed by only 0.279. We will report the multi-seed mean from now on.
- **Line endings.** The CSV committed to GitHub uses LF line endings while the original download uses CRLF, so their raw MD5 hashes differ (9b2b50f4... vs ee29c9ee...). The content is identical and all results are unchanged, so the checksum now normalises line endings.

- **Validation vs. CV.** They differ by up to 0.037 (decision tree, full set: 0.602 vs 0.639). The validation set has only 664 rows (119 Enrolled), so gaps of a few hundredths are within noise (CV sd 0.011-0.026).
- **Implementation checks.** The sklearn default max_iter=100 gives the same score (no convergence warning). Fitting the scaler and encoder on all rows, a leakage error, changes macro-F1 by only -0.002; the effect is small, but we keep fitting on training data only. The CV-tuned C equals the default, so "tuned" logistic regression is the same model.

## 4. What the results support, and what they do not

- Logistic regression beats the best naive benchmark by 0.338 macro-F1 on the early set, and Dropout recall is 0.714, so our proposal criterion is met on validation data.
- Most of the signal is available after semester 1: using semester 2 adds +0.054 macro-F1 on validation (+0.038 in CV).
- Enrolled is the weak class (F1 0.38 early, 0.49 full). On the early set, of 119 validation Enrolled students 47 are predicted Graduate and 34 Dropout. Accuracy (0.736) hides this; macro-F1 does not.
- Not supported: that logistic regression beats trees in general (the depth-5 tree is untuned and there is one split), any test-set claim (unused), or any causal claim.

**Baseline by group** (logistic regression, early features; out-of-fold predictions on train + validation, n = 3,760; groups from the group EDA, coded per the dataset descriptor; `results/subgroup_baseline.csv`):

| Group | n | Dropout rate | Dropout recall | Dropout precision | Macro-F1 |
|---|---|---|---|---|---|
| All | 3,760 | 0.321 | 0.745 | 0.777 | 0.660 |
| Female / Male | 2,448 / 1,312 | 0.254 / 0.446 | 0.699 / 0.793 | 0.774 / 0.780 | 0.663 / 0.641 |
| No scholarship / Scholarship | 2,824 / 936 | 0.385 / 0.129 | 0.778 / **0.446** | 0.787 / 0.651 | 0.663 / 0.548 |
| Fees up to date / **not** up to date | 3,314 / 446 | 0.248 / 0.863 | **0.633** / **0.984** | 0.710 / 0.892 | 0.635 / 0.491 |
| Age < 25 / >= 25 at enrollment | 2,802 / 958 | 0.243 / 0.550 | 0.641 / 0.879 | 0.706 / 0.859 | 0.642 / 0.645 |

- The baseline finds almost every dropout among students whose fees are unpaid (recall 0.98) but only 63% of those whose fees are paid, so much of its Dropout detection comes from financial variables, which we test in Milestone 5. It also misses most dropouts among scholarship holders (0.45). These are descriptive gaps in one dataset (small groups, e.g. 93 international students, are noisy), not evidence of causes.

## 5. Start / Stop / Continue

| Start | Stop | Continue |
|---|---|---|
| • Tuning the tree and random forest with the same CV effort as logistic regression (M4/M5).<br>• Trying class weighting and error analysis for the Enrolled class.<br>• Reporting the naive benchmark as a multi-seed mean.<br>• Small, frequent commits and one pull request plus one review per member; issues for each task; logging every run in `docs/EXPERIMENT_LOG.md`.<br>• Reading Martins et al. (2021, 2023) on early/phased prediction for this dataset to position our early-vs-full comparison.<br>• Fixing coding labels found in code review of the EDA notebook (descriptor: Gender 1 = male, attendance 1 = daytime). | • Judging models on a single validation score.<br>• Treating the depth-5 tree as a tuned model.<br>• Bulk-uploading work near the deadline (the repo history so far is a few large commits) and starting milestones late (M3 was submitted after its due date). | • The group EDA by gender, scholarship, tuition, debtor, displacement and attendance, extended with a subgroup check.<br>• Fixed seed, stored split, pipeline-only preprocessing and a pinned environment.<br>• Macro-F1 as primary metric with per-class recall.<br>• Early vs. full feature comparison and the leakage check on financial variables.<br>• Keeping the test set untouched until the final evaluation. |

**References.** [1] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting student dropout and academic success," *Data*, vol. 7, no. 11, p. 146, 2022, doi: 10.3390/data7110146. [2] N. T. Vo, Q. N. Le, and H. Le, "Artificial intelligence-driven early prediction of student dropout and academic outcomes in higher education," *EAI Endorsed Trans. Ind. Netw. Intell. Syst.*, vol. 13, no. 1, 2026, doi: 10.4108/eetinis.131.11758. [3] M. V. Martins, L. Baptista, J. Machado, and V. Realinho, "Multi-class phased prediction of academic performance and dropout in higher education," *Appl. Sci.*, vol. 13, no. 8, pp. 4702-4716, 2023 (cited by [2]; not yet read). [4] M. V. Martins, D. Tolledo, J. Machado, L. M. T. Baptista, and V. Realinho, "Early prediction of student's performance in higher education: a case study," *Adv. Intell. Syst. Comput.*, vol. 1365, pp. 166-175, 2021 (introductory paper listed in the repo README). [5] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825-2830, 2011.